SPAGHETTILAB | ONE BOARD, MANY POSSIBILITIES

Connect a circuit, choose how each port behaves, and explore what happens.

CONNECTIONS & POWER

OPTIONAL ESP32 MODULE

ESP32-S3 | Optional wireless module
Adds Wi-Fi, Bluetooth and ESP32 project support.

Adds Wi-Fi, Bluetooth and support for ESP32 projects.

DIRECT ACCESS

Connects accessories directly to the optional ESP32.

MAIN POWER

TPS562201 | Main power supply
Creates the power used by the board.

Turns the incoming power into the supply used by the board.

PORT LIGHTS

Shows the state of each port with coloured light.

EXPAND | ESP32 GPIO

J5: ESP32 + AG32 expansion

ESP GPIO + Qwiic + 4 AG32 signals. 3.3 V only.

GPIO35–37 reserved for PSRAM.

GPIO39–42 share JTAG.

External SPI on J5: 19=SCLK,20=MOSI,21=MISO,4=CS(GPIO11). Shared bus with AG32, separate CS. AG spare pins remain on TP42–44.

Connects extra modules through a flat cable at the board edge.

OPTIONAL ACCESSORY LINKS

Connects displays, cameras and other supported add-ons.

BOARD-TO-BOARD LINK

G3VM | Network end switch
Enables the end-of-network connection when needed.

Passes power and communication to neighbouring boards.

QWIIC ACCESSORIES

Connects compatible sensors with a small cable.

LIVE VIEW | SIX PORT VOLTAGES

Native ADC inputs remain on PORT_A0–A5.

Current is measured separately by INA226.

No external ADC, DAC, reference or op-amps.

NETWORK CONNECTION

SN65HVD230 | Network interface
Connects this board to the shared CAN network.

Lets this board exchange messages with the other boards.

USB CONNECTION

Connects the board to a computer.

OPTION | ESP32-ONLY CONTROLLER

Uses the ESP32 to configure the ports when AG32 is not fitted.

SIGNAL CONNECTIONS

U12 / U13 | Bidirectional signal switches
Connect or disconnect the ESP32 from the experiment ports.

Connects or separates the ESP32 signals from the experiment.

Existing circuit preserved. Tester features require firmware and prototype validation; no new tester components were added.

SIX EXPERIMENT PORTS

Gives each port a signal, power or ground connection.

Q3–Q8 | Electronic power switches
Give a port power or ground when selected.

CONTROL & LAB TOOLS

U14A–U14G are seven sections of ONE physical AG32 chip.

PORT SUPERVISOR

AG32 section A | Experiment signals
Reads the ports and handles digital signals.

Observes the ports and manages digital signals for the experiment.

PORT MODE CONTROL

AG32 section B | Port settings
Chooses signal, power or ground for each port.

Chooses how each experiment port is connected.

SUPERVISOR START-UP

20 MHz | Controller clock
Keeps the controller running at a steady pace.

AG32 section F | Power and start-up
Starts the controller and provides programming access.

Powers, starts and provides access to the board controller.

COMMUNICATION HUB

AG32 section C | Board communication
Exchanges data with the computer and other boards.

Links the controller to USB, the network and the optional ESP32.

CURRENT MEASUREMENT

AG32 section D | Current scan control
Chooses which port is measured.

4051 | First measurement selector
Connects one side of the selected current sensor path.

4051 | Second measurement selector
Connects the other side of the same sensor path.

INA226 | Current sensor
Measures the current of the selected port.

INA226 uses shared I2C IO4/IO5 (0x40); ESP32 and AG32 can configure/read it. Qwiic and FPC share the same bus.

Current mux control: GPIO6=S0, GPIO12=S1, GPIO41=S2, GPIO42=INHIBIT. Both CPUs MUST use open-drain outputs; inactive owner releases all four.

Inhibit first, select channel, wait for settling, then start a fresh INA conversion. This does not make ADC/DAC functions independent of AG32.

Selects one port at a time and measures the current flowing through it.

LOCAL POWER SUPPORT

Keeps the controller supply steady while it works.

SCAN | SMALL TEST SIGNALS

Internal DAC -> R77 -> selectable port. Read both sides of R77 with internal ADC.

Scan: MOSFETs OFF, digital drivers Hi-Z, ESP switches OPEN, R77 bypass OFF.

A return path is required. Occupancy detection is assisted, not guaranteed.

Applies a gentle test signal and watches how each connection responds.

ESP32 PROGRAMS THE AG32

Loads the board controller without an external programmer.

GPIO47 TX / GPIO48 RX | GPIO15 BOOT0 | GPIO16 RESET (open-drain)

AG UART0: TX pin68 / RX pin69. BOOT1 stays LOW via R84.

Recovery: keep ESP32 pins Hi-Z; use UART pads or the existing JTAG pads.

AG Base: U14 fitted, Lab components omitted, R56–R61 = 0R.

ESP Base: fit U25 and the dual-controller selectors. U25 is required in every automatically configurable variant.

Lab: U14 + measurement components fitted. USB remains direct to ESP32.

DUAL CONTROLLER — LED, PORT CONFIGURATION AND CAN

Selects the active controller. ESP32 works without AG32; AG32 works without ESP32. When both are fitted, AG32 takes control after a firmware handshake.

AG_READY: LOW selects ESP, HIGH selects AG. U25 alone drives all MOSFET gates and ISO0–3 via shared open-drain I2C (IO4/IO5).

Firmware: initialize safe outputs, release shared I2C, then hand over. Never drive the same bus from two masters without arbitration. No live power-gating support.

Single-page schematic. Functional descriptions are not performance specifications.

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